

Tinotenda Musikavanhu

Frontend Engineer • AI Integration • Systems Design

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🌐 Portfolio • [LinkedIn](#) • [GitHub](#)

Professional Summary

My full name is Tinotenda Musikavanhu, but professionally, I prefer to be addressed as Tino. My passion for software development emerged when I aspired to establish my own business. In this endeavor, I learned how to develop iOS applications and collaborated with a team of skilled professionals, including a senior engineer. This experience served as my initial exposure to the tech industry and sparked my curiosity to explore further. Over time, I immersed myself in multiple programming languages, and I was fortunate enough to share my knowledge with a vast audience of over 30,000 followers on social media platforms. What began as simple projects utilizing online resources transformed into personally gratifying work, and I even ventured into the realm of tutoring.

Overall, my journey into software development has been one of self-discovery, learning, and growth. I am confident that my experience, coupled with my passion for technology, has provided me with a solid foundation to excel in this field.

Education

Southern Methodist University

2016 – 2020

Computer Science

Emphasis: Systems Engineering & Artificial Intelligence

Location: Dallas, TX

Harvard University Extension School

Ongoing

Bachelor of Arts - Computer science

An intensive and comprehensive course covering a wide range of computer science topics. Acquired expertise in multiple programming languages, including JavaScript, SQL, and C++, and gained hands-on experience in software development, algorithms, data structures, and artificial intelligence. Developed strong analytical and problem-solving skills, enabling the ability to analyze complex problems and develop effective solutions. Participated in collaborative projects and worked with diverse teams, fostering excellent communication and teamwork abilities. Completed the program with a solid understanding of computer science principles and practices, ready to apply this knowledge to real-world challenges.

Professional Experience

Software Engineer
Safehaven

- Develop, maintain, and enhance Safehaven’s software products using JavaScript and other relevant technologies.
- Collaborated with cross-functional teams including designers, product managers, and other developers to implement new features and functionality.
- Wrote clean, efficient, and maintainable code that adheres to industry standards and best practices.
- Conducted code reviews and provide constructive feedback to other team members to ensure high-quality codebase.
- Ensure that software products are tested thoroughly and perform optimally across various platforms and browsers.
- Stayed up-to-date with the latest trends and advancements in JavaScript and related technologies, and incorporate them into the product development process.
- Participated in the development of technical documentation, user manuals, and other related materials.
- Troubleshoot and resolve technical issues reported by users, and provide timely support as needed.

Software Engineer Prototech

- Experienced Database Analyst and Developer with a proven track record of designing, developing, and maintaining complex databases for various applications. Adept at working collaboratively with cross-functional teams to understand business requirements and ensuring that database solutions meet those requirements. Possesses a strong ability to analyze and forecast potential database concerns, developing effective solutions and streamlining processes. Exceptional communication skills and proficiency in multiple programming languages including Angular, SQL, and Swift. Demonstrated success in enhancing client response time and delivering products ahead of schedule.
- Analyzed and Forecasted Database concerns for the NFL’s Employee management database on site at the Super Bowl. Executed more than 5,000 employee verifications, which Streamlined to 15% of all attendance to be assigned perfectly.
- Supported a new enterprise application in the Angular framework for risk management. Redesigned 75% of the existing database structure; Later shaped the product delivery to be assigned to the client two days earlier.
- Streamlined current bugs of employee management application directly with clients. Client response time and remedy now enhanced to be 20% faster with new hours of operation.

Technical Skills

Frontend & Web

- React, Next.js, TypeScript
- Framer Motion, Tailwind CSS
- Responsive Design, Web Perf

- Vanilla CSS, HTML5, SVG

AI & Machine Learning

- PyTorch, HuggingFace Transformers
- Latent Space & Diffusion Models

- LLM Integration (LM Studio, OpenAI)
- Reinforcement Learning

Backend & Systems

- Node.js, REST APIs, WebSockets
- Distributed Systems Architecture
- Real-time Data Pipelines

- Cyber-Physical Resilience

Tools & Platforms

- Git, GitHub, Vercel, Netlify
- Apple MPS / GPU Training
- Figma, System Design
- Zenodo Open Science

Selected Projects

AI Gameplay Automation System (Sims 4)

Python, React, Computer Vision

Autonomous AI agent that plays and manages a Sims 4 household using computer vision, decision trees, and reinforcement learning. Features a real-time React monitoring dashboard with live game-state telemetry.

GenesisTrader

Next.js, TypeScript, Coinbase API

AI-powered cryptocurrency trading system with predictive analytics, real-time market intelligence, and a high-performance Next.js frontend. Integrates local LLM reasoning for trade signal generation.

AI Systems Research — Latent Planning + Autoregressive Gen

PyTorch, HuggingFace, MPS

Independent research training custom transformer architectures to explore latent planning as a mechanism for improved autoregressive text and image generation quality.

Selected Publications

Full list available at: zenodo.org/Musikavanhu

- **Musikavanhu, T.** (2025). “SolarTherm-CarbCap (STCC): Integrated Solar-Thermal Carbon Capture for Climate Remediation.” DOI: [10.5281/zenodo.15711985](https://doi.org/10.5281/zenodo.15711985)
- **Musikavanhu, T.** (2025). “Thermodynamic Validation of Anthropogenic Climate Acceleration” DOI: [10.5281/zenodo.16422926](https://doi.org/10.5281/zenodo.16422926)
- **Musikavanhu, T.** (2025). “Quantum Zeta Field Theory: Unifying Framework for Quantum Fields and Mathematical Structures.”

Awards & Recognition

- **Open Science Impact** (2024–2025): 500+ research downloads across 64+ publications on Zenodo
- **NASA Recognition** (2025): Acknowledged by former NASA Administrator Dan Goldin for research contributions
- **High-Stakes Ops** (2019–2025): Zero-downtime delivery for Super Bowl-scale event deployments at ProtaTECH

“Building the future — one component, one model, one breakthrough at a time.”